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DEREK EUBANKS, PE, HFDP, HBDP, CPD
SENIOR MECHANICAL ENGINEER, PE | HEALTHCARE & MISSION-CRITICAL MEP | COMPUTATIONAL ENGINEERING & APPLIED AI

PROFESSIONAL SUMMARY

Licensed Mechanical Engineer (PE – CA, GA, TN) with 13+ years of experience designing HVAC and hydronic infrastructure for mission critical facilities including hyperscale data centers, hospitals, and advanced manufacturing environments. Expert in Revit-based system design, hydronic plants, and high-performance ventilation systems. Currently pursuing a M.S. in Computer Science with an Artificial Intelligence specialization at Georgia Tech - applying software development, machine learning, and computational methods to engineering design and optimization.

SKILLS & SOFTWARE

- Autodesk Revit Expert
- Bluebeam Revu
- Microsoft Office Suite
- Python, C# (Revit API), VBA, Java
- D3.js, Matplotlib, Seaborn, Pandas, NumPy, SciPy, Scikit-learn
- Trane Trace 3D Plus; Carrier HAP
- High-performance Mission-Critical and Healthcare HVAC & Plumbing & Medical Gas Design
- Hydronic waterside plant design
- QA/QC for HVAC, plumbing, and medical gas drawing markups

EXPERIENCE

MECHANICAL ENGINEER PE III • ENFRA • JUNE 2026 – PRESENT; REMOTE

- Lead mechanical design and coordination for a new 8-story healthcare facility, developing HVAC systems and reviewing multidisciplinary construction documents for code compliance, constructability, and system performance.
- Perform detailed QA/QC reviews of HVAC, hydronic, controls, and equipment documentation using ASHRAE 170, ASHRAE 62.1, FGI, IMC, and project-specific design criteria.
- Research and develop AI-assisted and Revit-based QA/QC workflows to automate equipment, schedule, model, and drawing consistency checks.

SENIOR MECHANICAL ENGINEER • SPACEX • JULY 2025 – JAN 2026; ON-SITE

- Engineered variable-primary-flow heat-pump chiller plants, HVAC systems, and hydronic distribution for advanced manufacturing facilities using Trace 3D Plus and Revit.
- Developed complete 3D Revit MEP models, equipment layouts, hydronic piping, controls schematics, and sequences of operation from field-verified existing conditions.
- Delivered complete demolition and new-construction mechanical drawing packages—from laser-verified field data to permit-ready documentation—in ≤1 week.

SENIOR ASSOCIATE/SENIOR MECHANICAL ENGINEER • SYSKA HENNESSEY GROUP • JAN 2024 – MARCH 2025; ON-SITE

- Led mechanical design for a 48 MW Tier III hyperscale data center, developing N+1 chilled-water and HVAC infrastructure supporting data halls and critical electrical spaces.
- Engineered and coordinated a ~21,800 GPM chilled-water system, 40+ air-cooled chillers, CRAH galleries, and three mechanical pump rooms in Revit.
- Directed mechanical design development, equipment selection, load analysis, and interdisciplinary BIM coordination through permit-level construction documentation.

SENIOR MECHANICAL ENGINEER • JACOBS • OCT 2022 – DEC 2023; REMOTE

- Designed HVAC and hydronic systems for Tesla and Rivian advanced-manufacturing facilities, performing load analysis and full 3D Revit coordination.
- Engineered and modeled multiple 40,000+ CFM chilled-water AHUs, associated ductwork and piping, heat-pump systems, and low-dewpoint environmental-control systems.
- Coordinated multidisciplinary design development while delegating calculations and design tasks to junior engineers.

*Full portfolio, code, and project writeups available via GitHub and LinkedIn

ADDITIONAL ENGINEERING EXPERIENCE:

- **SENIOR MECHANICAL ENGINEER • MAZZETTI • APR 2020 – MAY 2023; REMOTE**
  - Designed HVAC, plumbing, and medical-gas systems for major healthcare facilities using Revit, Trace 3D Plus, and detailed psychrometric/hydronic analysis.
  - Developed VBA-based engineering tools and detailed BAS controls schematics, points lists, and sequences of operation.
- **MECHANICAL DISCIPLINE DIRECTOR • GRAY AES • AUG 2018 – MAR 2020; ON-SITE**
  - Served as Mechanical EOR and discipline leader, directing engineers/designers, managing project assignments, and leading HVAC/plumbing design for healthcare and industrial clients.
  - Led proposals, project management, performance reviews, and mechanical discipline coordination.

EDUCATION:

- **MASTER OF SCIENCE IN COMPUTER SCIENCE • GEORGIA INSTITUTE OF TECHNOLOGY • 2025–2028**
  - Specialization: Artificial Intelligence
  - GPA: 3.67/4.0
  - Completed: CS 6603, CSE 6242, CS 6310
  - In Progress: CS 6601 Artificial Intelligence
- **BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING • GEORGIA INSTITUTE OF TECHNOLOGY • 2007–2013**
  - Dean's List (2) semesters; Merit list (2) semesters
  - Math GPA: 3.68/4.0 (7 courses); Computer Science GPA: 3.5/4.0 (2 courses)
  - HOPE scholarship & Federal PELL Grant recipient for maintaining >= 3.0 GPA

GRADUATE AI, ML, & COMPUTATIONAL PROJECTS:

- Adversarial Search Agent | Python | CS 6601 Artificial Intelligence
  - Developed a game-playing agent using Minimax, Alpha-Beta Pruning, Iterative Deepening, heuristic evaluation, and depth-limited adversarial search.
  - Implemented recursive MAX/MIN logic and runtime-aware search optimization; earned 100/100.
- Graph & Heuristic Search Algorithms | Python | CS 6601 Artificial Intelligence
  - Implemented BFS, UCS, A*, Bidirectional UCS, Bidirectional A*, and Tridirectional Search for weighted graph traversal and optimal-path discovery.
  - Applied priority queues, admissible heuristics, frontier management, and path-cost optimization.
- Airline Delay Root Cause Explorer | Python / Scikit-learn | CSE 6242 Data & Visual Analytics
  - Built an end-to-end ML pipeline using 400k–800k+ BTS/NOAA records, including feature engineering, preprocessing, modeling, and evaluation.
  - Compared Logistic Regression, Random Forest, and Gradient Boosting models; achieved ROC-AUC ~0.70 and performed feature-importance analysis.
- Algorithmic Fairness Analysis | Python / Scikit-learn | CS 6603 AI, Ethics, and Society
  - Applied Logistic Regression and Random Forest models to 54k+ records, evaluating bias using Statistical Parity Difference and Equal Opportunity Difference.
  - Evaluated reweighting-based bias mitigation while preserving predictive performance.

LICENSES & CERTIFICATIONS:

- Professional Engineer (PE Licenses): California (#42960), Georgia (#PE042696), Tennessee (#130418)
- MyNCEES PE-verified engineering employment history since 2013 – eligible to pursue comity licensure in other states.
- ASHRAE Healthcare Facility Design Professional (HFDP)
- ASHRAE High-Performance Building Design Professional (HBDP)
- ASPE Certified in Plumbing Design (CPD)
- Autodesk Certified Professional in Revit for Mechanical Design